

VIRTUAL BATTERY FACTORY

DESIGN DELIVERY PACKAGE

Case: t1_r3 — next-generation pure electric sedan battery
ED $\geq$ 392.61 Wh/kg · 4C no plating · T_{max} $\leq$ 60 °C · 4.7 V overcharge without TR
Generation date: 2026-08-26 · prepared/reviewed/approved: (blank for manual signing)

Document number: VBF-EXPT1R3-IDX-01

Cover Information

Field	Value
Case name	t1_r3 (next-generation pure electric sedan battery: ED $\geq$ 392.61 Wh/kg, 4C fast charge without plating, T _{max} $\leq$ 60 °C, overcharge 4.7 V without thermal runaway)
Numbering scheme	`VBF-<CASE-ID-UPPERCASE>`-<DOC-CODE>`-<SEQ-NO>`; case ID `exp/t1_r3` → `EXPT1R3`
Generation date	2026-08-26
Prepared by	(blank)
Reviewed by	(blank)
Approved by	(blank)

Document Code Reference

Document code	Meaning	Corresponding file
DS	Specification	`design_spec.md`
BOM	Bill of Materials	`bom.xlsx`
DSH	Datasheet	`datasheet.md`
CALC	Calculation sheet	`calc.xlsx`
DVPR	Design verification report	`dvpr.md`
DFMEA	Failure analysis	`dfmea.md`
CAD	Structure model	not produced (optional; headless session — no user clarification requesting it; honestly skipped)
IDX	Delivery index	`delivery_index.md`

File List

File name	Number	Format	Source description
design_spec.md	VBF-EXPT1R3-DS-01	md	generated per `references/deliverable-design-spec.md`; all values from parameter set + `r5_v11_*.json` outputs
design_spec.pdf	VBF-EXPT1R3-DS-01	pdf	md→pdf release export via reportlab
report.html	VBF-EXPT1R3-DS-02	html	`bda render --case-dir runs/exp/t1_r3` seven-section log report (ancillary to DS)
bom.xlsx	VBF-EXPT1R3-BOM-01	xlsx	generated per `references/deliverable-bom.md` via openpyxl
bom.pdf	VBF-EXPT1R3-BOM-01	pdf	xlsx→pdf release export via reportlab
datasheet.md	VBF-EXPT1R3-DSH-01	md	generated per `references/deliverable-datasheet.md`
datasheet.pdf	VBF-EXPT1R3-DSH-01	pdf	md→pdf release export via reportlab
calc.xlsx	VBF-EXPT1R3-CALC-01	xlsx	generated per `references/deliverable-calc-sheet.md` via openpyxl (5 sheets)

File name	Number	Format	Source description
calc.pdf	VBF-EXPT1R3-CALC-01	pdf	xlsx→pdf release export via reportlab
dvpr.md	VBF-EXPT1R3-DVPR-01	md	generated per `references/deliverable-dvpr.md` (virtual-test version)
dvpr.pdf	VBF-EXPT1R3-DVPR-01	pdf	md→pdf release export via reportlab
dfmea.md	VBF-EXPT1R3-DFMEA-01	md	generated per `references/deliverable-dfmea.md` (qualitative version)
dfmea.pdf	VBF-EXPT1R3-DFMEA-01	pdf	md→pdf release export via reportlab
delivery_index.md	VBF-EXPT1R3-IDX-01	md	this file (cover + controlled file list)
delivery_index.pdf	VBF-EXPT1R3-IDX-01	pdf	md→pdf release export via reportlab (blueprint-style cover)